

STATISTICS

PART 6: Estimators and Confidence Interval (for single population)

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Estimators and Confidence Interval

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INFERENTIAL STATISTICS

- **Inferential statistics** is a process that infers results from a population using sampling. This process is conventionally divided into two parts:

Inferential Statistics

Estimation

In the estimation process, we **infer numerical results about population parameters**.

Hypothesis Testing

We choose one **hypothesis** from **two opposite hypotheses** related to the population parameters or population distribution.

- A sample statistic that is used for the estimation of an unknown population parameter is called an **estimator**, its value is called an **estimated value**.

Population Parameter (θ)	Estimator $\hat{\theta}$	Estimate value $\hat{\theta}$
Mean (μ)	$\bar{X}$	$\bar{x}$
Variance (σ^2)	S^2	s^2
Standard Deviation (σ)	S	s
Proportion (p)	$\hat{P}$	$\hat{p}$

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The Properties of Estimators

1. Unbiased Estimators: A statistic $\hat{\theta}$ is an **unbiased estimator** of the population parameter θ if and only if

$$E(\hat{\theta}) = \theta.$$

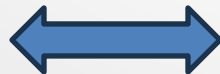
Examples:

$$E(\bar{X}) = \mu$$



The sample mean $\bar{X}$ is an unbiased estimator of the population mean μ .

$$E(\hat{p}) = p$$



The sample proportion $\hat{p}$ is an unbiased estimator of the population parameter p .

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Theorem: If S^2 is the variance of a random sample selected from an infinite population with a variance σ^2 , then

$$E(S^2) = \sigma^2.$$

The sample variance S^2 is an unbiased estimator of the population variance σ^2 .

Proof:

$$\begin{aligned}
 E(S^2) &= E \left[\frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2 \right] = E \left[\frac{1}{n-1} \sum_{i=1}^n [(X_i - \mu) - (\bar{X} - \mu)]^2 \right] \\
 &= E \left[\frac{1}{n-1} \sum_{i=1}^n [(X_i - \mu)^2 - 2(X_i - \mu)(\bar{X} - \mu) + (\bar{X} - \mu)^2] \right] \\
 &= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X} - \mu) \underbrace{\sum_{i=1}^n (X_i - \mu)}_{n\bar{X} - n\mu} + n(\bar{X} - \mu)^2 \right) \right] \\
 &= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right) \right]
 \end{aligned}$$

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$$\begin{aligned}
 E(S^2) &= E \left[\frac{1}{n-1} \left(\sum_{i=1}^n (X_i - \mu)^2 - n(\bar{X} - \mu)^2 \right) \right] \\
 &= \frac{1}{n-1} \left(\sum_{i=1}^n \underbrace{E[(X_i - \mu)^2]}_{\sigma^2} - n \underbrace{E[(\bar{X} - \mu)^2]}_{\sigma_{\bar{x}}^2} \right) \\
 &= \frac{1}{n-1} \left(\sum_{i=1}^n \sigma^2 - n \underbrace{\sigma_{\bar{x}}^2}_{\sigma^2/n} \right) \\
 &= \frac{1}{n-1} (n\sigma^2 - \sigma^2) = \sigma^2
 \end{aligned}$$

As shown in the proof, the sample variance S^2 is an unbiased estimator when the denominator of its formula is defined by $n-1$ instead of n .

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The Properties of Estimators

2. Efficiency: If $E(\hat{\theta}_1) = \theta$, $E(\hat{\theta}_2) = \theta$ and $\text{Var}(\hat{\theta}_1) < \text{Var}(\hat{\theta}_2)$, then it is said that $\hat{\theta}_1$ is more efficient than $\hat{\theta}_2$.

$$\text{Relative Efficiency of } \hat{\theta}_1 \text{ with respect to } \hat{\theta}_2 = \frac{\text{Var}(\hat{\theta}_2)}{\text{Var}(\hat{\theta}_1)}$$

If

$$E(\hat{\theta}) = \theta \text{ and } \text{Var}(\hat{\theta}) = \frac{1}{nE \left[\left(\frac{\partial \ln f(x)}{\partial \theta} \right)^2 \right]}$$

then $\hat{\theta}$ is said to be unbiased estimator with minimum variance of θ or the most efficient estimator of θ .

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Example 1: Show that $\bar{X}$ is the unbiased estimator with minimum variance of μ which is the mean of a normal distributed population.

Solution:

$$E(\bar{X}) = \mu \longrightarrow \text{The condition 1 is satisfied.}$$

Since the probability function of normal distribution is defined by

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty,$$

it follows that

$$\begin{aligned} \ln f(x) &= -\ln(\sigma\sqrt{2\pi}) - \frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 \\ \frac{\partial \ln f(x)}{\partial \mu} &= \frac{1}{\sigma}\left(\frac{x-\mu}{\sigma}\right) \end{aligned}$$

$$E\left[\left(\frac{\partial \ln f(x)}{\partial \mu}\right)^2\right] = E\left[\left(\frac{1}{\sigma}\left(\frac{x-\mu}{\sigma}\right)\right)^2\right] = \frac{1}{\sigma^4} E[(x-\mu)^2] = \frac{1}{\sigma^2}$$

$\underbrace{\hspace{10em}}_{\sigma^2}$

The condition 2 is satisfied.

$$\begin{aligned} \frac{1}{nE\left[\left(\frac{\partial \ln f(x)}{\partial \mu}\right)^2\right]} &= \frac{1}{n\frac{1}{\sigma^2}} = \frac{\sigma^2}{n} \quad \text{and} \quad \text{Var}(\bar{X}) = \sigma_{\bar{x}}^2 = \frac{\sigma^2}{n} \\ \Rightarrow \text{Var}(\bar{X}) &= \frac{1}{nE\left[\left(\frac{\partial \ln f(x)}{\partial \mu}\right)^2\right]} \end{aligned}$$

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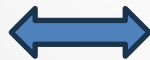
Example 2: Let X_1 and X_2 represent the elements of two samples that are randomly selected from a population whose mean is μ and variance is σ^2 . Four estimators of μ are given below:

$$\hat{\mu}_1 = \frac{1}{2}X_1 + \frac{1}{2}X_2, \quad \hat{\mu}_2 = \frac{2}{5}X_1 + \frac{3}{5}X_2, \quad \hat{\mu}_3 = \frac{1}{5}X_1 + \frac{2}{5}X_2, \quad \hat{\mu}_4 = \frac{1}{4}X_1 + \frac{3}{4}X_2$$

- Which estimators are unbiased estimators?
- Which estimator is the most effective?
- Compute the relative efficiency of $\hat{\mu}_1$ with respect to $\hat{\mu}_2$ and interpret the results.

Solution:

$$E(\bar{X}) = \mu$$



The sample mean $\bar{X}$ is an unbiased estimator of the population mean μ .

$$E(\hat{\mu}_1) = E\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \frac{1}{2}\underbrace{E(X_1)}_{\mu} + \frac{1}{2}\underbrace{E(X_2)}_{\mu} = \mu \quad \Rightarrow \hat{\mu}_1 \text{ is an unbiased estimator of } \mu.$$

$$E(\hat{\mu}_2) = E\left(\frac{2}{5}X_1 + \frac{3}{5}X_2\right) = \frac{2}{5}\underbrace{E(X_1)}_{\mu} + \frac{3}{5}\underbrace{E(X_2)}_{\mu} = \mu \quad \Rightarrow \hat{\mu}_2 \text{ is an unbiased estimator of } \mu.$$

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$$E(\hat{\mu}_3) = E\left(\frac{1}{5}X_1 + \frac{2}{5}X_2\right) = \frac{1}{5}\underbrace{E(X_1)}_{\mu} + \frac{2}{5}\underbrace{E(X_2)}_{\mu} \neq \mu \quad \Rightarrow \hat{\mu}_3 \text{ is not an unbiased estimator of } \mu.$$

$$E(\hat{\mu}_4) = E\left(\frac{1}{4}X_1 + \frac{3}{4}X_2\right) = \frac{1}{4}\underbrace{E(X_1)}_{\mu} + \frac{3}{4}\underbrace{E(X_2)}_{\mu} = \mu \quad \Rightarrow \hat{\mu}_4 \text{ is an unbiased estimator of } \mu.$$

$$\text{var}(\hat{\mu}_1) = \text{var}\left(\frac{1}{2}X_1 + \frac{1}{2}X_2\right) = \left(\frac{1}{2}\right)^2 \underbrace{\text{var}(X_1)}_{\sigma^2} + \left(\frac{1}{2}\right)^2 \underbrace{\text{var}(X_2)}_{\sigma^2} = \frac{1}{2}\sigma^2$$

$$\text{var}(\hat{\mu}_2) = \text{var}\left(\frac{2}{5}X_1 + \frac{3}{5}X_2\right) = \left(\frac{2}{5}\right)^2 \underbrace{\text{var}(X_1)}_{\sigma^2} + \left(\frac{3}{5}\right)^2 \underbrace{\text{var}(X_2)}_{\sigma^2} = \frac{13}{25}\sigma^2$$

$$\text{var}(\hat{\mu}_4) = \text{var}\left(\frac{1}{4}X_1 + \frac{3}{4}X_2\right) = \left(\frac{1}{4}\right)^2 \underbrace{\text{var}(X_1)}_{\sigma^2} + \left(\frac{3}{4}\right)^2 \underbrace{\text{var}(X_2)}_{\sigma^2} = \frac{10}{16}\sigma^2$$

$$\text{Var}(\hat{\mu}_1) < \text{Var}(\hat{\mu}_3) < \text{Var}(\hat{\mu}_4)$$



$\hat{\mu}_1$ is more effective than other estimators.

$$\text{Relative Efficiency of } \hat{\mu}_1 \text{ with respect to } \hat{\mu}_2 = \frac{\text{Var}(\hat{\mu}_2)}{\text{Var}(\hat{\mu}_1)} = \frac{13\sigma^2/25}{\sigma^2/2} = 1.04 \quad \Rightarrow \text{if we use } \hat{\mu}_2 \text{ instead of } \hat{\mu}_1, 1.04 \text{ times larger sample size we need.}$$

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Estimation

Point Estimation

Interval Estimation

If the value of the estimator is accepted as an estimate of the population parameter, this process is called **point estimation**.

Example 3: A computer firm produces graphic cards. 5 Graphic cards are randomly selected. If the lasting times of these cards are 8, 11, 5, 7 and 19 months, compute point estimations for the mean and variance of lasting times of all produced graphic cards in the firm.

Solution:

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n} = \frac{50}{5} = 10 \quad \Rightarrow \quad \bar{x} = \mu = 10$$
$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = 30 \quad \Rightarrow \quad s^2 = \sigma^2 = 30$$

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Estimation

Point Estimation

Interval Estimation

The probability of observing a single value in the continuous distributions such as normal, t or χ^2 distributions, is theoretically zero. More clearly, if θ is a population parameter, $\hat{\theta}$ is an estimator and $\hat{\theta}$ is the estimated value (point estimate) of the population parameter, then $P(\theta = \hat{\theta}) = 0$. Thus, using a point estimation alone provides **no confidence** (zero confidence).

To increase confidence level, we can form an error interval around $\hat{\theta}$ by assuming that the estimation includes an error (e), namely $(\hat{\theta} - e, \hat{\theta} + e)$. In this case,

$$P(\hat{\theta} - e \leq \theta \leq \hat{\theta} + e) = 1 - \alpha$$

where $1 - \alpha$ is the probability that θ lies within the interval $(\hat{\theta} - e, \hat{\theta} + e)$. The interval $\hat{\theta} - e \leq \theta \leq \hat{\theta} + e$ is the **100(1 - α)% confidence interval for the population parameter θ** . The value $1 - \alpha$ is the **confidence level for this interval estimation**.

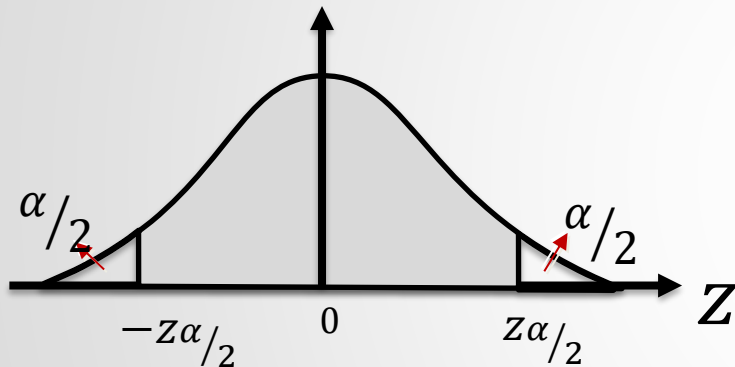
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The Confidence Interval for the Population Mean

For Normal Distributed Populations with Known Variance



$$\Rightarrow P\left(-z_{\alpha/2} \leq \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

To extract μ in inequality, ... $\Rightarrow P\left(-z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \bar{x} - \mu \leq z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$

$$\Rightarrow P\left(-\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq -\mu \leq -\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha$$

100(1 - α)% confidence
interval for μ

$$\Rightarrow \left(\underbrace{\bar{x} - z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}_e, \underbrace{\bar{x} + z_{\alpha/2} \frac{\sigma}{\sqrt{n}}}_e\right)$$

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The Confidence Interval for the Population Mean

for normal distributed
population with
unknown variance

$$\Rightarrow P\left(-t_{\alpha/2, n-1} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}} \leq t_{\alpha/2, n-1}\right) = 1 - \alpha \Rightarrow \left(\underbrace{\bar{x} - t_{\alpha/2, n-1} S/\sqrt{n}}_e, \underbrace{\bar{x} + t_{\alpha/2, n-1} S/\sqrt{n}}_e\right)$$

for unknown
distributed
population with
known variance
($n \geq 30$)

$$\Rightarrow P\left(-z_{\alpha/2} \leq \frac{\bar{x} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha \Rightarrow \left(\underbrace{\bar{x} - z_{\alpha/2} \sigma/\sqrt{n}}_e, \underbrace{\bar{x} + z_{\alpha/2} \sigma/\sqrt{n}}_e\right)$$

for unknown
distributed
population with
unknown variance
($n \geq 30$)

$$\Rightarrow P\left(-z_{\alpha/2} \leq \frac{\bar{x} - \mu}{S/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha \Rightarrow \left(\underbrace{\bar{x} - z_{\alpha/2} S/\sqrt{n}}_e, \underbrace{\bar{x} + z_{\alpha/2} S/\sqrt{n}}_e\right)$$

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Example 4: It is known that the IQ scores of bachelor students in Turkey are distributed normally with a variance of 160. To estimate the mean of the distribution, 12 students are randomly selected, and their IQ scores are measured as follows:

103, 96, 112, 130, 111, 125, 89, 100, 104, 117, 99, 118

Construct a 95% confidence interval for the mean of all bachelor students in Turkey.

$$\begin{aligned} n &= 12 & \bar{X} &= \frac{\sum_{i=1}^{12} X_i}{12} = 108.67 & \alpha &= 1 - 0.95 = 0.05 \\ \sigma &= \sqrt{160} & & & z_{\alpha/2} &= z_{0.025} = 1.96 \end{aligned}$$

$$\left(\bar{x} - z_{\alpha/2} \sigma / \sqrt{n}, \bar{x} + z_{\alpha/2} \sigma / \sqrt{n} \right)$$

$$\Rightarrow \left(108.67 - 1.96 \sqrt{160} / \sqrt{12}, 108.67 + 1.96 \sqrt{160} / \sqrt{12} \right)$$

$$\Rightarrow (101.51, 115.83)$$

Thus, we can claim that the population mean μ is between the interval [101.51, 115.83] with a confidence of 95%.

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Example 5: Solve the previous problem in case that the population variance is unknown.

$$n = 12 \quad \bar{X} = \frac{\sum_{i=1}^{12} X_i}{12} = 108.67$$

$$S = \sqrt{\frac{\sum_{i=1}^{12} (X_i - \bar{X})^2}{12 - 1}} = 12.3$$

$$\alpha = 1 - 0.95 = 0.05$$

$$t_{\alpha/2, n-1} = t_{0.025, 11} = 2.201$$

$$\left(\bar{x} - t_{\alpha/2, n-1} S / \sqrt{n}, \bar{x} + t_{\alpha/2, n-1} S / \sqrt{n} \right)$$

$$\Rightarrow \left(108.67 - 2.201 \cdot 12.3 / \sqrt{12}, 108.67 + 2.201 \cdot 12.3 / \sqrt{12} \right)$$

$$\Rightarrow (100.85, 116.49)$$

Thus, we can claim that the population mean μ is between the interval [100.85, 116.49] with a confidence of 95%.

Estimators and Confidence Interval

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Example 6: To estimate the fuel consumption of a specific model of a car, 40 cars have been tested and it has been detected that the mean is 8.1 l/km and the standard deviation is 0.67 l/km. Construct a 95% confidence interval for the exact mean of all cars of the specific model.

$$n = 40$$

$$\bar{X} = 8.1$$

$$S = 0.67$$

$$\alpha = 1 - 0.95 = 0.05$$

$$z_{\alpha/2} = z_{0.025} = 1.96$$

$$\left(\bar{x} - z_{\alpha/2} \frac{S}{\sqrt{n}}, \bar{x} + z_{\alpha/2} \frac{S}{\sqrt{n}} \right)$$

$$\Rightarrow \left(8.1 - 1.96 \frac{0.67}{\sqrt{40}}, 8.1 + 1.96 \frac{0.67}{\sqrt{40}} \right)$$

$$\Rightarrow (7.89, 8.31)$$

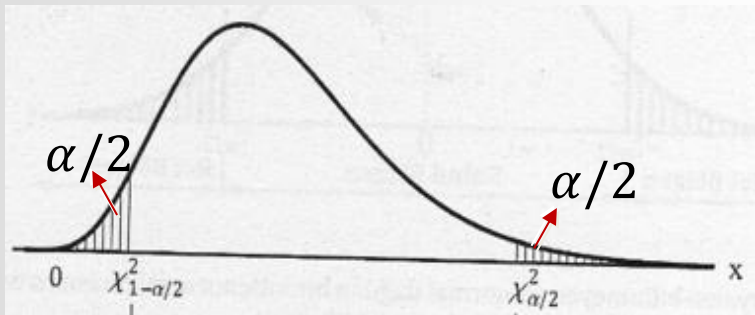
Thus, we can claim that the population mean μ is between the interval [7.89, 8.31] with a confidence of 95%.

Estimators and Confidence Interval

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The Confidence Interval for the Population Variance

For Normal Distributed Populations:



$$\Rightarrow P\left(\chi^2_{1-\frac{\alpha}{2}, n-1} \leq \frac{(n-1)S^2}{\sigma^2} \leq \chi^2_{\frac{\alpha}{2}, n-1}\right) = 1 - \alpha$$

To extract σ^2 in the inequality, ...

$$\Rightarrow P\left(\frac{1}{\chi^2_{\frac{\alpha}{2}, n-1}} \leq \frac{\sigma^2}{(n-1)S^2} \leq \frac{1}{\chi^2_{1-\frac{\alpha}{2}, n-1}}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}} \leq \sigma^2 \leq \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}}\right) = 1 - \alpha$$

100(1 - α)% confidence interval for σ^2

$$\Rightarrow \left[\frac{(n-1)S^2}{\chi^2_{\frac{\alpha}{2}, n-1}}, \frac{(n-1)S^2}{\chi^2_{1-\frac{\alpha}{2}, n-1}} \right]$$

Estimators and Confidence Interval

Statistics

Example 7: The standard deviation of the fuel consumption of an engine that has been worked 16 times is 2.2 gallons. Construct a 99% confidence interval for the variance of the exact fuel consumption of the engine. (Assume that the population variance has a normal distribution.)

$$n = 16$$

$$S = 2.2$$

$$\alpha = 1 - 0.99 = 0.01$$

$$\chi_{\frac{\alpha}{2}, n-1}^2 = \chi_{0.005, 15}^2 = 32.801$$

$$\chi_{1-\frac{\alpha}{2}, n-1}^2 = \chi_{0.995, 15}^2 = 4.601$$

$$\left[\frac{(n-1)S^2}{\chi_{\frac{\alpha}{2}, n-1}^2}, \frac{(n-1)S^2}{\chi_{1-\frac{\alpha}{2}, n-1}^2} \right]$$

$$\Rightarrow \left[\frac{(16-1)2.2^2}{32.801}, \frac{(16-1)2.2^2}{4.601} \right]$$

$$\Rightarrow (2.21, 15.78)$$

Thus, we can claim that the population variance σ^2 is between the interval $[2.21, 15.78]$ with a confidence of 99%.

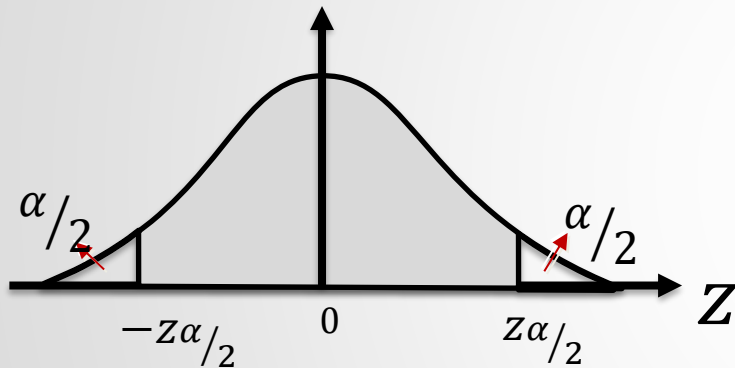
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The Confidence Interval for the Population Proportion

For $n \geq 30$



$$P\left(-z_{\alpha/2} \leq \frac{\hat{p} - p}{\sqrt{\hat{p}\hat{q}/n}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

To extract p in inequality, ...



$$P\left(-z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n} \leq \hat{p} - p \leq z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}\right) = 1 - \alpha$$



$$P\left(-\hat{p} - z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n} \leq -p \leq -\hat{p} + z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}\right) = 1 - \alpha$$



$$P\left(\hat{p} - z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n} \leq p \leq \hat{p} + z_{\alpha/2}\sqrt{\hat{p}\hat{q}/n}\right) = 1 - \alpha$$

100(1 - α)% confidence
interval for p



$$\left[\underbrace{\hat{p} - \frac{z_{\alpha/2}}{2}\sqrt{\hat{p}\hat{q}/n}}_e, \underbrace{\hat{p} + \frac{z_{\alpha/2}}{2}\sqrt{\hat{p}\hat{q}/n}}_e \right]$$

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Example 8: 136 of 400 experimental people who were vaccinated for influenza have caught the disease. Construct a 95% confidence interval for the proportion of people who caught the disease among all vaccinated people.

$$n = 400 \quad \hat{p} = \frac{136}{400} = 0.34$$
$$\hat{q} = 1 - \hat{p} = 0.66$$

$$\alpha = 1 - 0.95 = 0.05$$
$$z_{\alpha/2} = z_{0.025} = 1.96$$

$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\hat{p}\hat{q}/n}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\hat{p}\hat{q}/n} \right]$$

$$\Rightarrow \left[0.34 - 1.96 \sqrt{(0.34)(0.66)/400}, 0.34 + 1.96 \sqrt{(0.34)(0.66)/400} \right]$$

$$\Rightarrow (0.294, 0.386)$$

Thus, we can claim that the population proportion p is between the interval $[0.294, 0.386]$ with a confidence of 95%.

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In Case of Selection of a Sample without Replacement

When a sample is selected without replacement from a population of size N , and $n/N > 0.05$, the correction factor should be considered. In this case, since

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \quad \text{or} \quad \sigma_{\bar{x}} = \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \quad \sigma_{\hat{p}} = \sqrt{\frac{pq}{n}} \sqrt{\frac{N-n}{N-1}}$$

the confidence intervals for the means and proportion become as follows:

for normally distributed population
with known variance

for any population with known
variance, $n > 30$

$$\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

for normally distributed population
with unknown variance



$$\left[\bar{x} - t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + t_{\frac{\alpha}{2}, n-1} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

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for any population with unknown
variance, $n > 30$



$$\left[\bar{x} - z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}}, \bar{x} + z_{\frac{\alpha}{2}} \frac{S}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

$n > 30$



$$\left[\hat{p} - z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}}, \hat{p} + z_{\frac{\alpha}{2}} \sqrt{\frac{\hat{p}\hat{q}}{n}} \sqrt{\frac{N-n}{N-1}} \right]$$

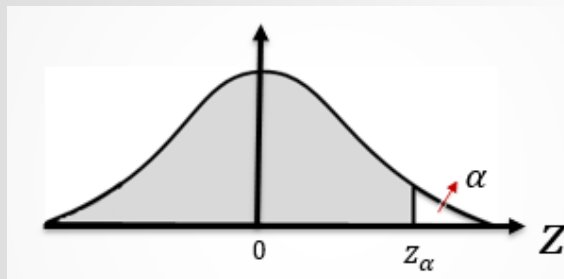
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ONE-SIDED CONFIDENCE INTERVAL

Although using two-sided confidence intervals are commonly used, it is also possible to construct a one-sided confidence interval. Consider the z distribution given below:



$$P(Z < Z_{\alpha}) = 1 - \alpha$$

If the population has a normal distribution with a known variance, ...

$$\Rightarrow P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} < Z_{\alpha}\right) = 1 - \alpha$$

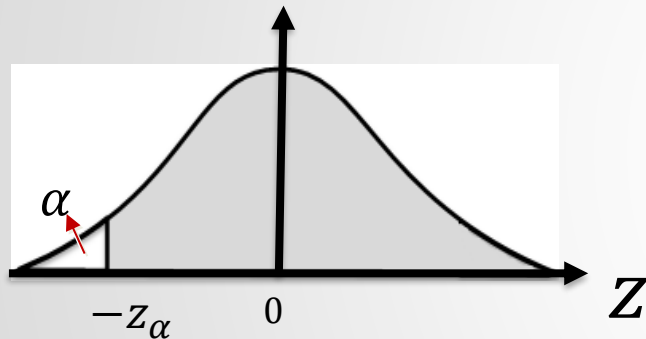
$$\Rightarrow P\left(\mu > \bar{X} - Z_{\alpha} \sigma / \sqrt{n}\right) = 1 - \alpha \Rightarrow \mu \in \left[\overbrace{\bar{X} - Z_{\alpha} \sigma / \sqrt{n}}^{\text{Lower bound}}, \infty \right]$$

One-sided (lower) confidence interval

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For the left side,



$$\Rightarrow P(Z > -Z_\alpha) = 1 - \alpha$$

If the population has a normal distribution with a known variance, ...

$$\Rightarrow P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > -Z_\alpha\right) = 1 - \alpha$$

upper bound

$$\Rightarrow P\left(\mu < \bar{X} + Z_\alpha \sigma/\sqrt{n}\right) = 1 - \alpha \Rightarrow \mu \in \left[-\infty, \overbrace{\bar{X} + Z_\alpha \sigma/\sqrt{n}}^{\text{upper bound}}\right] \quad \text{One-sided (upper) confidence interval}$$

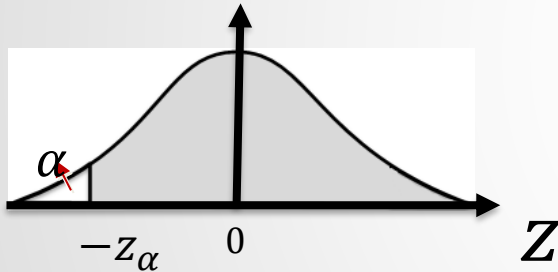
Similarly, one-sided confidence intervals can be derived for variances, proportions, and population means, even when the populations are non-normally distributed and/or have unknown variances.

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Example 9: In a psychological test experiment, the reaction times of 25 randomly selected subjects to a particular stimulus were measured. Past experiences show that the reaction time is normally distributed with a variance of 4 sec. The mean reaction time of 25 subjects is 6.2 sec. Compute the 95% upper bound for the mean reaction time.



$$P\left(\frac{\bar{X} - \mu}{\sigma/\sqrt{n}} > -Z_{\alpha}\right) = 1 - \alpha$$

$$\Rightarrow P\left(\mu < \bar{X} + Z_{\alpha} \sigma/\sqrt{n}\right) = 1 - \alpha$$

upper bound $\Rightarrow \bar{X} + Z_{0.05} \sigma/\sqrt{n}$

$$= 6.2 + (1.645) \sqrt{4}/\sqrt{25}$$
$$= 6.858 \text{ sec.}$$

$$\Rightarrow \mu \in [-\infty, 6.858] \text{ for } \alpha=0.05$$